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24. The discrete turn: A reconsideration of architecture's ontology

Mollie Claypool

Time, in other words, reappeared in the world as something real, as a destabilizing but creative milieu; it was seen to suffuse everything, to bear each thing along, generating it and degenerating it in the process. Soon there was no escaping the fact that transformation and novelty were the irreducible qualities that any theory of form would need to confront.

—Sanford Kwinter¹

There is a rich tradition in architecture for technological advancements to influence the way that architects design the built environment. For example, the Industrial Revolution had a huge impact on how, and what, buildings were made of, with the invention of mass-produced steel and concrete, and the post-World War II period of rapid expansion of housing stock was due to technological advancements made during the war, such as the development of mass-standardised production methods. Digital technologies have also transformed the very landscape of the way the world is experienced, from electronic devices and cars that have dramatically increased in quality and efficiency to the use of smart gadgets, machine learning and IoT devices in homes, offices, civic, cultural and public spaces. The digital economy is the underpinning for powerful platforms such as Airbnb, Google, Amazon, Facebook and Uber that have shifted the way that our cities are inhabited, products are consumed and people's data are utilised.

Architecture has been eagerly complicit in the adoption of digital technologies since the early 1990s, experimenting with the potential of digital tools adopted from the manufacturing and film industries, as well as innovations and developments in fields outside of architecture such as biology, philosophy and chemistry. This can be seen in the work of architects such as Greg Lynn's early simulation studies of pedestrians, cars

and other forms of movement for the design of the Port Authority Bus Terminal in New York City (1994) or Reiser + Umemoto's Watergarden for Jeff Kipnis (1997). Designs could be simulated, transcribed and translated in a multitude of ways using digital design tools, and given layers of complexity through the embedding of the notion of 'parameters' inscribed within the design process.

This shift towards the digitisation of architectural design practice has been further supported by the amplification of the importance of studying the processes of designing, or what is referred to commonly as 'design research'. Architecture could be viewed not merely as a teleology but as a practice which viewed research as an intrinsic part of the processes of design. Therefore, architecture could no longer be viewed only as a static, inert object, post-rationalised in terms of its meaning, but had the possibility to be a dynamic, fluxing, shifting, complex process of design, through and from which meaning emerged.

This period of expansive change within architecture and design is of course only possible to understand with some retrospection. In architecture, this period has been historicised by architecture historian Mario Carpo first through the notion of the first digital turn, which roughly took place from 1992 until 2012.² In the first digital turn, Carpo wrote that because electronic, for example digital, technologies were drastically changing the landscape of production across almost all industries, societies and economies, 'so much was changing, that some architects started to think that design should change too'. What became clear to architects was that these technological changes would extend from the amplification of the cyber-virtual environment of the Internet to the built environment's physical production.³ Yet, in the first digital turn, what has become evident is that the virtual data that were created were being realised with production methods that required a degree of post-rationalisation. Otherwise, the complex forms that were generated were extremely difficult to produce. As a result, the new forms that were being created were being fit into earlier, and old, production paradigms that created long, complex, expensive production chains for architectural realisation.

The Big Data revolution of the 2010s of increased computational power transformed all industries. Post 2012, Carpo has outlined what he refers to as the second digital turn, which presumably extends to the moment we are in today. According to Carpo, the second digital turn has at its core a computational approach which enables digital tools to 'follow a different, nonhuman, postscientific method' that is characterised by letting 'computers solve problems in their own way' rather than in the way humans historically have done, developing a 'new kind of science' that is

‘begetting a new way of thinking’.⁴ Data, the time required to process it and the computational systems used to process it have become ever-more accessible to designers to search for possible design outcomes or solutions that can mirror their physical counterparts. In addition, processes used to produce design outcomes can be coded into the design space and used alongside other parameters to predict and accommodate the changing behaviour of material, technology and people.

A return to architecture itself: The discrete

This shift between the first and second digital turn is a shift from emphasising process over formal appearance, from discontinuity to continuity between the virtual and physical manifestations of architecture, from systems of construction which are analogue and traditional to ones which are digital, agile and dynamic, from understanding matter as inert to understanding it as active. How was this manifested at the start of this shift? How, retrospectively, was the groundwork for the second digital turn laid? And how has this influenced architectural understandings of the relationships between information, material resources, society and space? This chapter documents this shift through early projects of the second digital turn that laid the groundwork for what today is referred to as the discrete.

The discrete is an architectural approach informed by mereology – or the study of part-to-whole relationships – that completely rethinks the role of the digital in architecture, in terms of tectonics, space, materiality and environment. The discrete also takes a sociopolitical position critical of the generation of architects of the first digital turn, arguing that in addition to the digital forms of production – both virtual and physical production – that are available today, architecture itself – its parts, assembly, tectonics, materials – also needs to be digital. This has been done by understanding a set of architectural elements as digital data, like the 1s and 0s of computer code, not entirely dissimilar to Lego, which can be combined and recombined in multiple but finite orientations and used in many different ways across different scales. Discrete architecture is argued by others, such as Gilles Retsin and Jose Sanchez, and me in a recent *Architectural Design* (April 2019) issue as being a more accessible, versatile, open-ended and participatory approach to architecture.⁵

As discrete architectural parts are self-similar in tectonics and geometry, discrete parts are able to harness aspects of mass production and mass standardisation in production in both manufacturing and assembly

techniques. This makes them prime for use with advanced automated fabrication technologies such as computer-numerically controlled (CNC) milling and robotic assembly. These technologies need to be materially and resource light, much more so than traditional architectural parts which are usually materially and manufacturing resource intensive. As such, the discrete is a move towards a better understanding of architecture itself.

Architecture itself: Matter and machines

Perhaps it is best to understand this shifting towards discrete thinking through an understanding of the relationship between design and representation. In the late 1990s, around the same time as the height of the first digital turn, the hypothesising of the relationship of the drawing to building was done by the late architect and educator Robin Evans. Evans wrote in his seminal essay 'Translations from Drawing to Building' (1997) that the substratum across which translation between drawing and building occurs is a space of opportunity – for invention, manipulation, accidents to occur, very much in the same way it occurs in language. One just has to look at the English edition translations of texts of post-structuralist French philosophers to realise that this is the case – that in any other language, meaning becomes construed, manipulated and, oftentimes, transformed. This of course can be interpreted as a positive attribute to the process of translation – as surprises can lead to innovation, novelty and the breaking of existing paradigms. What Evans recognised, however, is that within architecture, the language of representation that found a place of prominence was the two-dimensional drawing. While the drawing could become as far away as possible from the actual thing itself – the object of architecture or a realised physical manifestation of it – and maintain its importance to the discipline, what this has resulted in in terms of material practice is that the two-dimensional drawing had found a place in architecture where its representation of a thing became more important than the 'properties' of architecture itself.⁶

This thinking existed in parallel to materialist practices in architecture that historically fell primarily into one of two groups throughout the twentieth century: that which imposed lack of material difference in favour of an ideal, and that which equalised material difference in order to determine form. The modernist project is the main protagonist in both of these coteries, but one can look back as far as the pyramids, as Le Corbusier did in *Towards An Architecture* (1923) almost a hundred

years ago, to recognise they both have existed throughout the history of architecture. Achieving form was, by and large, a process of forcing inert material to become something the architect wanted it to be. Material was often treated as a passive mono-material, active only when manipulated to achieve an idealised form. For example, in the work of Louis Kahn, brick was viewed as a homogenous series of idealised elements, relatively undifferentiated in its accumulations. However, this is not limited to the work of modernists. One only has to look at the many built works of Zaha Hadid Architects (ZHA) to understand that this phenomenon is still at work today where the post-rationalisation of matter must occur in order to realise the geometric complexity of ZHA buildings.

As the three-dimensional model slowly became the output of many architecture practices and construction firms, object-orientated design (OOD) became more common. In 2012, 71 per cent of design and construction companies in North America utilised building information modelling software.⁷ This emphasis on the three-dimensional modelling software transformed the two-dimensional drawing into a three-dimensional object embedded with specifications and data for the design of a building. The difference between drawing and building therefore became lessened as both the fields of architecture and construction were able to interpret and utilise the same three-dimensional model in real time.

At the time, this process was limited to the design process, rather than being extended to methods of fabrication, construction or inhabitation, and resulted in the object of architecture being considered inert and static once constructed. But what would the implications be if there was an ontology for architecture which enabled it to be adaptable, flexible, changeable, in real time or throughout a longer life cycle? What if the very matter of architecture did not require homogenisation due to the imposition of equalising of materialist and structuralist practices with existing – and old – methods of construction or means of inhabitation? And what would it mean for architecture to utilise technologies in a way to enable this shift?

The discipline has a now well-established interest in industrial digital fabrication technologies of production such as CNC milling, three-dimensional 3D printing, laser cutting and robotic fabrication, amongst others. Although these are interlinked but outside the direct realm of operation of most architects of the late twentieth century, automated machines have a clear history within architecture. One only has to look at Sigfried Giedion's *Mechanisation Takes Command* (1948) to see that this is the case. However, through object-orientated thinking, only in recent years have these technologies become a means of developing an approach

to architecture's matter that takes into account heterogeneity, difference and variability as a means of breaking the dominant paradigms of imposing or equalising of the twentieth century. While at the start of the first digital turn these technologies mainly existed within the silo of academic institutions or industry, it is rare today in many parts of the world to find a school of architecture that does not have a digital fabrication workshop.

Additive and subtractive processes of manufacturing, when utilised alongside digital simulation software and a critical reconsideration of the architectural object, enabled Evans's reading of the relationship between drawing and building to come back into question. The shift of the architect away from the drawing and towards architecture itself has revolutionised the translation of the work of architects to the construction industry. The introduction of digital fabrication methods means that architects can now ask architectural elements – its matter – what they want to be, rather than imposing upon them a prescribed outcome. The three-dimensional model can become part of a feedback loop between physical behaviour, digital simulation and desired outcomes, taking into account that the space between the model and construction allows space for design ingenuity and opportunity, as well as constructive, productive and meaningful failure. This also means that architects can consider the emergent behaviours of material and tectonic arrangements or patterns. Coming back to Kahn's use of brick, what this kind of practice allows in architecture is a dissociation with the notion that a material's purposiveness lies in its homogeneous application to an architectural object or building.

Looking briefly at four projects that were developing in the early part of the second digital turn, they are certainly not definitive or exhaustive of the themes looked at during this period in digital design in architecture. Rather, these projects demonstrate methods used that can now be substantiated as initial moments of experimentation that led towards the more contemporary discrete approach embedded within Carpo's *Second Digital Turn*. Gilles Retsin and Isaie Bloch's proposal 'Karosta Kube' (2014) for the Homemade Dessert and Liepaja City Council, Latvia, competition drew on a novel non-linear interpretation of OOD that they have called Object-Orientated Eclecticism. Philippe Morel's (EZCT Architecture & Design Research) work (2012) using ultra-high-performance fibre-reinforced concrete (UHPFC) explored the potential of recursive lattice structures. Manuel Jiménez Garcia developed an open-source application called SoftModelling (2012–), drawing from democratic evolutions in computer science on the accessibility of code and an interest in engineered material practices. The final project of Research Cluster 4, The Bartlett

School of Architecture, UCL students Nan Jiang, Yiwei Wang, Zheeshan Ahmed and Yichao Chen (taught by Retsin and Jiménez Garcia) titled 'Space Wires' (2015) looked at how to optimise, in real time, a structural heterogeneous three-dimensional space frame, utilising robotic fabrication.

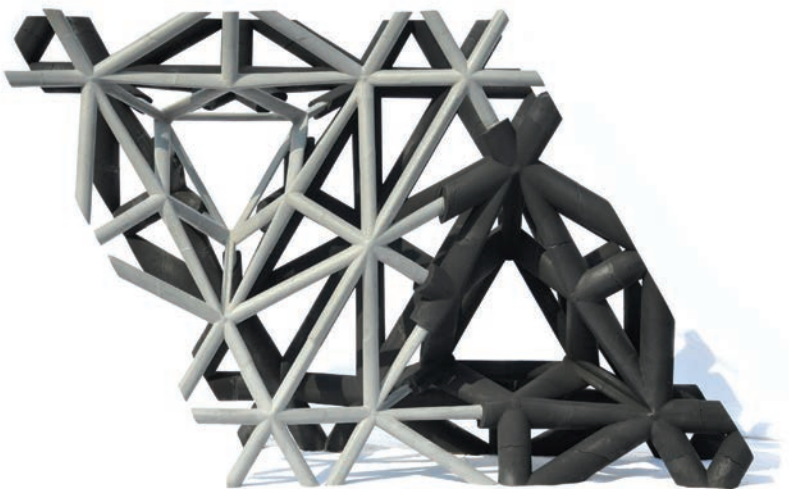
Discrete thinking: Four projects

Philippe Morel has used discrete mathematical models in architectural design research primarily through explorations into material behaviour. Morel's work into this area is not particularly recent, as he has had a well-established body of research into the topic since the mid-2000s with his work titled 'Computational Chair' (2006), but the piece that his practice EZCT Architecture & Design Research exhibited in the FRAC Centre exhibition 'Naturalising Architecture' (2013) in Orléans, France, titled 'Studies in Recursive Lattices' (2013), has become an emblem for the continued problem of homogeneity versus discreteness in architecture. Where Morel differs from the other projects included in this piece is in his recognition, at the time, of the very historicisation of this problem. Morel argues that it is due to our phenomenological perception of traditional building materials and material practices that we see it as homogenous, when in actuality, it is very much discrete – that is, bricks are distinctly separate elements. However, one still cannot ignore the fact that although bricks are components of the same thing, the use of the brick historically is treated in a continuous and homogenous way.

What Morel's 'Studies in Recursive Lattices' (2013) achieved is both continuity and discreteness in material efficiency, both geometrically and, as a result, structurally. There is continuity in terms of the material used throughout: UHPFC. This material has a degree of strength and ductility that far outweighs the normal concrete used ubiquitously worldwide in the construction industry. It was, however, still extremely expensive. By aiming to provide an alternative to the high cost of current strategies for the use of UHPFC, Morel and his collaborators worked on three-dimensionally modelling geometries derived from studies of recursive behaviour in biological lattice structures. They were then able to divide the lattice into structurally and geometrically optimised segments. This process used substantially less concrete material than in traditional concrete construction, as the three-dimensional prints of the segmented moulds were highly optimised and able to be tested digitally for their structural and geometric accuracy (figures 24.1 and 24.2) before being produced, thus saving both time and cost.



24.1 Philippe Morel/EZCT Architecture & Design Research, XTreeE, Space-Truss Prototype, 2012. The inverse of the structural lattice segments, three-dimensionally printed to produce precise sand moulds in which the ultra-high-performance fibre-reinforced concrete was cast. © Philippe Morel, EZCT Architecture & Design Research, XTreeE.



24.2 Philippe Morel/EZCT Architecture & Design Research, XTreeE, Space-Truss Prototype, 2012. Overall view of the 1:1 scale prototype, a section of a larger lattice structure that was envisioned. © Philippe Morel, EZCT Architecture & Design Research, XTreeE.

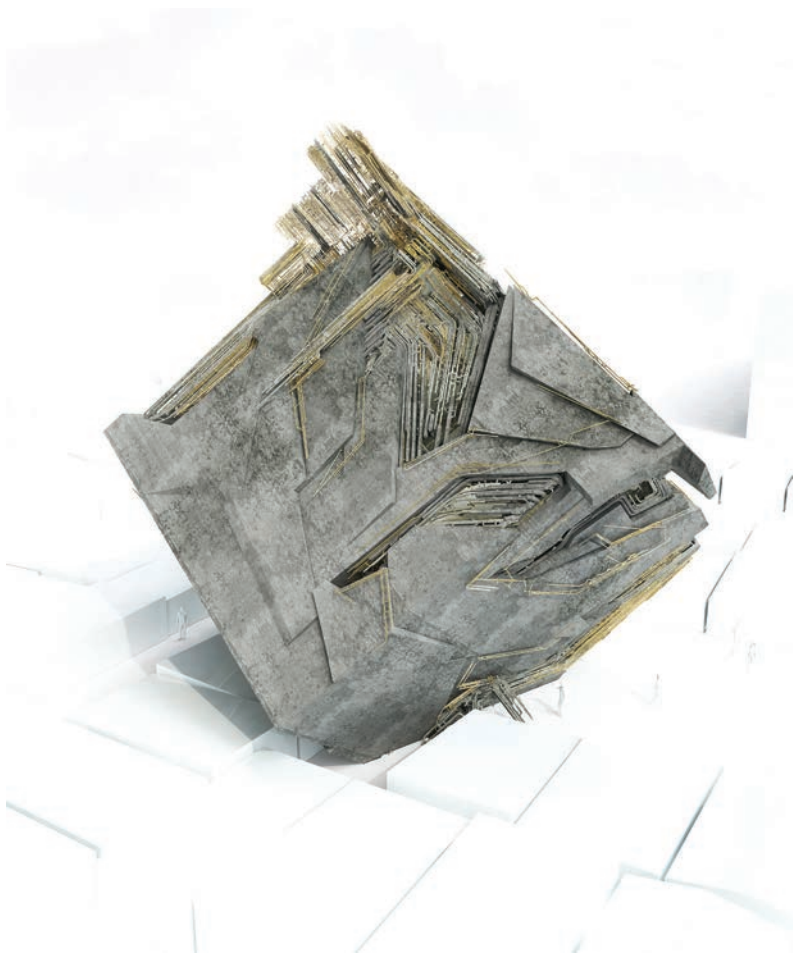
The potential of this method for the use of UHPFC is huge, as it produced topologically and geometrically complex results, both spatially and structurally, at a cost which in the future could be much lower than conventional concrete. The biggest problematic was, however, the relative slowness of the construction industry to utilise robotic fabrication (or

most forms of digital fabrication) on a large scale to produce structural lattices. However, Morel's project did take into account the necessity for a structural lattice to be discrete, meaning that on-site construction of segments of moulds into a continuous lattice was not, at the time, viewed as being far off from being a reality. Through this project, Morel demonstrated that the processes of design and production can exist in synthesis: virtual models and physical behaviour are the same.

The work of Gilles Retsin (Gilles Retsin Architecture) and Isaie Bloch (Eragatory) also took forward a similar topic that was explored by Morel. Yet, in the search for heterogeneity, Retsin and Bloch took the notion of discreteness from mathematics and applied it alongside the principle of irreducibility, or the notion that a unit is one that 'retains all the basic properties of the whole, and which cannot be further divided without losing them'.⁸ This concept can be brought closely to the concept of computational irreducibility described by Stephen Wolfram in *A New Kind of Science* (2002).⁹ Wolfram defined this term due to the lack of precise, formulaic mathematical description for many common systems in science, using cellular automata as an example.¹⁰

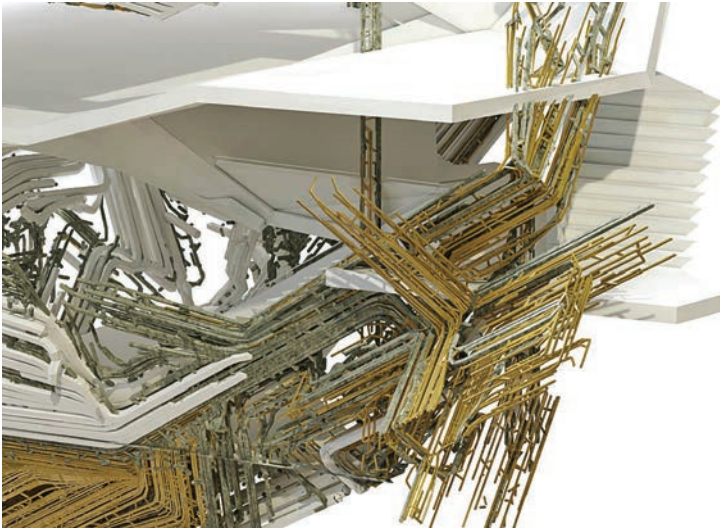
This idea is related in history to Thomas Kuhn's work in *The Structure of Scientific Revolutions* (1962). In this seminal and widely contested text, Kuhn wrote that within the practice of normal science, or puzzle-solving, scientific knowledge was gained in a linear, accumulative manner, a 'piecemeal process by which these items [of knowledge are] added singly and in combination'.¹¹ The process of normal science took place within widely established theories and practices. However, revolutions in science occurred if what Kuhn termed as a paradigm shift took place, when anomalies that arose through mistakes, omissions, miscalculations and failures highlighted inaccuracies in the everyday practice of science.

Furthermore, these inaccuracies highlighted the incompatibility of dominant methods of practice with novel technologies, or emerging contemporary modes of thinking and experimentation. Scientific revolutions for Kuhn could broadly be assimilated to Wolfram's notion of computational irreducibility. As Wolfram wrote, 'in effect there can be no way to predict how the system will behave except by going through almost as many steps of computation as the system itself'.¹² Novelty arises from these computationally irreducible processes. Retsin and Bloch took this notion and utilised it as a framework for design process in their proposal titled 'Karosta Kube' (2013) (figures 24.3 and 24.4), actively attempting to set up binary and dichotomous design processes that resulted in unpredictable tectonic behaviour.



24.3 Gilles Retsin/Gilles Retsin Architecture & Isaie Bloch/Eragatory, 'Karosta Kube', London, UK, 2013. Overall view of the front side and entrance of 'Karosta Kube' from a bird's-eye view as positioned on site in Liepaja, Latvia, showing two converging tectonic languages. © Gilles Retsin and Isaie Bloch.

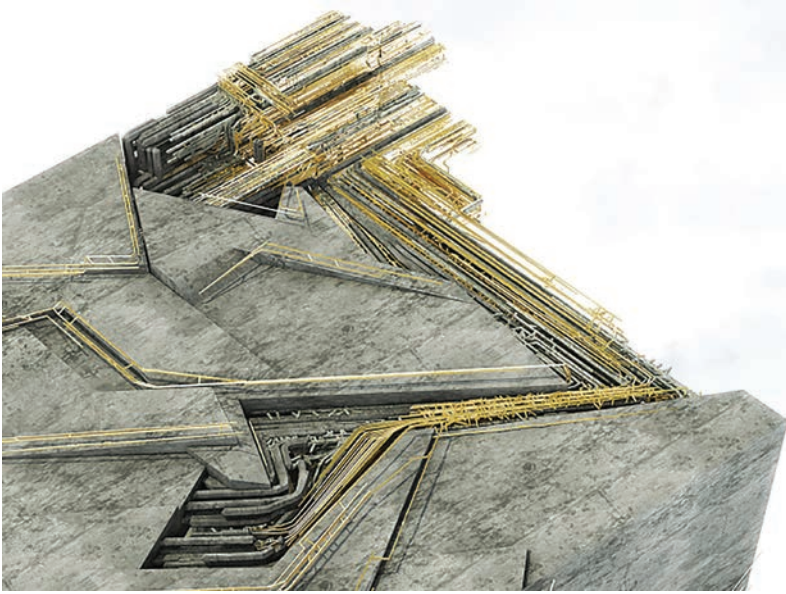
'Karosta Kube' is globally defined by two objects, one of which is more of what one would recognise as an 'object' than the other. There was a concrete block – a rotated, square shell – and then linear strands of steel rebar to which the term 'object' would not commonly be described. Each architect worked on only one object before attempting to combine them together. The concrete is defined as an inherently heterogenous structure, with a difference in density that is not continuous. The deposition of linear rebar material within and around the concrete is therefore dependent on the proximity of the rebar to this difference (figure 24.9). Where



24.4 Gilles Retsin/Gilles Retsin Architecture & Isaie Bloch/Eragatory, 'Karosta Kube', London, UK, 2013. Detail of 'Karosta Kube', showing two tectonic languages converging: steel rebar and concrete. © Gilles Retsin and Isaie Bloch.

there is more difference within the concrete, there is more articulation of rebar material. The result of this is a strange object which is not the result of any singular, traceable computational process. It is irreducible, as its properties emerged from the computational processes of articulating material behaviours and their tectonics (figure 24.5).

Moving on, we can see the impact of Frei Otto's experiments with catenary structures and varied material systems (soap bubbles etc.) from the 1960s to 1980s, as well as the research into the material behaviour of concrete catenary systems by Antoni Gaudi in Colonia Guell (1908–14) and Sagrada Familia (1882–1926) as a precedent for the work of Manuel Jiménez Garcia. Many pieces of digital modelling software emulate the analogue processes that these two architects and engineers worked with. However, as the use of physical digital simulation in architecture has exponentially increased, new problematics of utilising these tools have arisen. Advanced simulation software allowed, on the one hand, a more accurate understanding of material behaviour at an architectural scale and, on the other, a form-finding method. The potential of continual structural evaluation in form finding allowed for the morphology of an architectural system to be informed by physical laws instead of mathematical definitions in real time, enabling the evaluation of multiple iterations of the same system to happen simultaneously. Despite the ever-increasing familiarity of form-finding tools in architectural design practice, there



24.5 Gilles Retsin/Gilles Retsin Architecture & Isaie Bloch/Eragatory, 'Karosta Kube', London, UK, 2013. View of side of proposal where spatial elements emerge from the tectonic convergence of two design languages. © Gilles Retsin and Isaie Bloch.

were few attempts at this time to rethink or hybridise the more common digital design tools that architects utilised.

The application that Jiménez Garcia developed, called SoftModelling (figure 24.6), interrogated the way in which these tools are utilised, with a particular interest in how to incorporate concepts drawn from open-source models of computing and production into the software packages that architects used daily. These more generic software packages, which are heavily licensed and regulated, limited the development of open-source tools in architectural design research largely within the problem solving linked to specific architectural projects or problems. SoftModelling, on the other hand, is an open-source Java application developed not only to address a specific project, but also to cover the basic function of a digital design software, as its code was open source and easy to manipulate in order to create multiple versions by using processing as a framework. It is multi-scalar in its application, as it connected together two of the most-used design tools: poly-modelling and physical simulation. SoftModelling functioned differently from other software packages available, tying together physical behaviour and modelling into a digital feedback loop. To give an example, when modelling in Maya, the designer has to convert the model into a physical simulation but cannot operate



24.6 Manuel Jiménez García/MadMDesign, SoftModelling, 2012. View of the design interface simulating particle-spring systems and geometry simultaneously. © Manuel Jiménez García.

directly on the model when it is in the physical simulation. Kangaroo for Grasshopper works similarly, as you have to model the architectural object and then run the simulation, et cetera. Jiménez García improved on this problem in SoftModelling by seamlessly integrating between modelling and physics.

Most kinds of modelling software recomputed the order of edges when any mesh operation is given. This is why a two-step process was normally utilised, since the serial numbers of the particle springs will not match the new edges' serial numbers after this operation occurs. SoftModelling developed a strategy for each of the mesh operations in order to solve this. First, the app relocated the serial numbers of each edge on the mesh to maintain parity between the particle springs linked to them (figures 24.7 and 24.8). Then, instead of a recompilation of the particle-spring system, a detailed analysis of the mesh identified the parts that have been modified, without affecting the rest of the object.

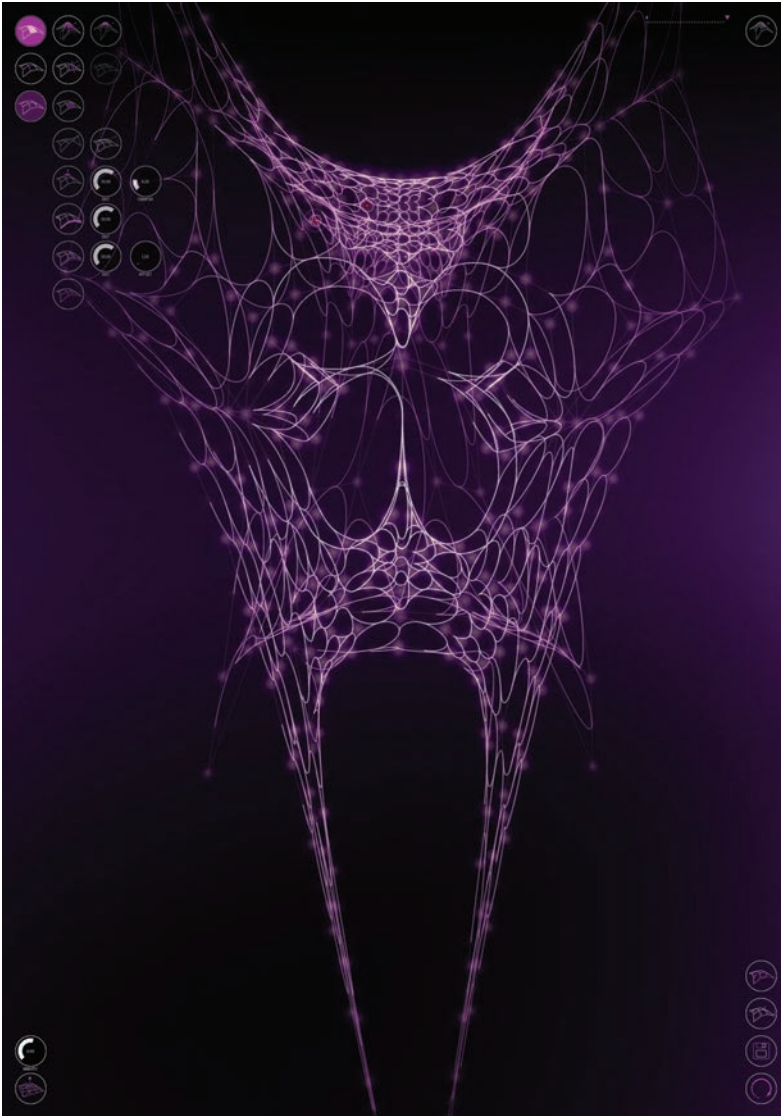
This process not only improved the efficiency of the physics simulation but also facilitated a seamless integration between modelling and simulation. The synchronisation of particles – vertices/springs to edges – enabled the constant updating of the positions of each part of the model. What one models is automatically physics, and vice versa. There was a continuous feedback between the physical behaviour of every



24.7 Manuel Jiménez García/MadMDesign, SoftModelling, 2012. View of the design interface simulating a catenary curve structure, side view. © Manuel Jiménez García.

particle spring of the three-dimensional mesh subdivision, as well as the variable scale and depth at every point, which led to an output that was both physically and geometrically precise. This improved flexibility for the designer, as one can modify and simulate simultaneously within a single piece of software, as well as edit the graphical user interface and the interface. It established an understanding that particle-spring systems can be used not only as a global framework but as a step-by-step transformative process for architectural design that is agile and flexible within an overarching framework for a design space that enables seamless feedback between software, architectural outcomes and the designer. This framework also enables scalability, as parameters can be adjusted according to material, tectonic, environmental or spatial constraints automatically within the design space.

The project 'Space Wires' by Nan Jiang, Yiwei Wang, Zheeshan Ahmed and Yichao Chen from Research Cluster 4 at the Bartlett School of Architecture, UCL, took a somewhat different approach, but utilised many of the concepts outlined above, including building open-source applications in order to marry structure and geometry more closely with construction and fabrication technologies. This is no coincidence, as they were taught by Jiménez García and Retsin at the Bartlett School of Architecture, UCL. The research cluster had, at the time, a particular

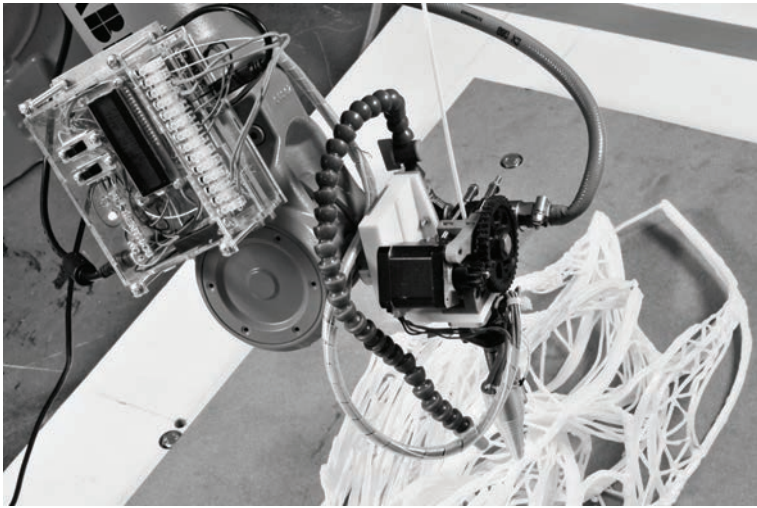


24.8 Manuel Jiménez García, MadMDesign, SoftModelling, 2012. Zoomed-in detail of the design interface showing systems interaction. © Manuel Jiménez García.

interest in the appropriation of three-dimensional printing and robotic fabrication technologies for multi-hierarchical and multi-material architectural design strategies. The aim was to discover material and fabrication anomalies in normative uses of both three-dimensional printing and robotics (figures 24.9 and 24.10), as well as traditional uses for materials such as concrete, clay and plastics, and utilise these anomalies



24.9 Nan Jiang, Yiwei Wang, Zheeshan Ahmed, Yichao Chen, Space Wires, Research Cluster 4, The Bartlett School of Architecture, UCL, 2014. Overall view of simulation of 1:1 physical prototype. © Research Cluster 4, The Bartlett School of Architecture, UCL.



24.10 Nan Jiang, Yiwei Wang, Zheeshan Ahmed, Yichao Chen, Space Wires, Research Cluster 4, The Bartlett School of Architecture, UCL, 2014. Printing a segment of the 1:1 prototype with custom extruder on industrial robot. © Research Cluster 4, The Bartlett School of Architecture, UCL.

as opportunities for innovation in regards to architectural tectonics. This project focused on the use of filament plastics.

‘Space Wires’ relates to the work on contour crafting by Behrokh Khoshnevis of University of Southern California, ‘D-Shape’ by Enrico Dini, ‘Mataerial’ by researchers Petr Novikov and Saša Jokić from

Barcelona's Institute for Advanced Architecture of Catalonia at Joris Laarman Lab and 'Chairs for Charity' by Dirk Vander Kooij. Through the invention of new technologies for robotic three-dimensional plastic filament extrusion – such as printing heads for materials that can have the potential for multiple extrusion geometries – the research was able to achieve heterogeneity in terms of both structural complexity and spatial complexity in both robotic fabrication and digital computation.

Where the group was most innovative is in their combination of two observations: first, that space-frame lattices bear loads much greater than their own self-weight; and second, that traditional three-dimensional printing technologies waste a lot of material. Oftentimes, much more waste is produced than utilised. They therefore utilised agent-based systems in combination with robotic plastic filament extrusion to generate structural data in real time, mimicking the geometric patterns of a space-frame lattice structure. The structural data and resultant behaviour of material is analysed in real time. Any anomalies, mistakes or failures that occurred in the output of the material system were fed back into the digital model, allowing for the output to be continuously updated.

The discrete turn

All of the projects described above attempted to bring together more closely the design process with construction through the utilisation of novel digital tools and fabrication technologies in combination with a critique of the ways previous generations of architects utilised these tools. In 'Space-Truss Prototype', discrete thinking was embedded in the tectonics of continuity that structural space-frame lattices require in order to cope with their structural load. Furthermore, the infrastructure for production considers material behaviour in relationship to cost and accommodates for this from the outset of the project, as it was a design parameter. Similarly, the kind of design-to-manufacturing system 'Space Wires' developed holds the potential to revolutionise construction techniques due to its ability to adjust to changes in tolerance due to site conditions such as weather, soil composition or the inaccuracy of the machine itself. It is a system that is able to be combinatorial, versatile and agile. It is also capable of achieving multi-hierarchical resolution of surfaces, utilising the same system but slightly modifying its printing pattern, the viscosity of the material being printed and the path of the robotic printing head. SoftModelling folds real parameters of multiple systems into a single virtual design and simulation space, allowing for

real-time feedback between systems, which is essential to discrete thinking today. 'Karosta Kube' demonstrates how aspects of the discrete – such as the notion of emergent behaviours or patterns through combinatorial thinking – can be played out through irreducible kits of parts and playing with architectural tectonics.

The properties of architecture itself take a place of prominence in each of these works. By revising an architectural objects ontology – the organisation of matter – the projects began to shift approaches in tectonics, fabrication, assembly and materiality. They aimed ultimately to improve accessibility, increasing and acknowledging that heterogeneity and efficiency in production and fabrication, in terms of both cost and labour, allowing architects to achieve a much greater specificity in the use of materials in design research. Heterogeneity is achieved not through what philosopher Manuel De Landa has recognised as the 'spontaneous generation of form', but through topological and tectonics difference and variability, best expressed by 'complex and variable behaviour' of discrete elements.¹³

What De Landa went on to note is that this can result in continuous variation, where variation in densities of material can result in a material performance that is heterogeneous, acknowledging that a 'single universal material is [not] good for all different kinds of structure'.¹⁴ When matter is considered to be in a state of continuous variation – that is, when it is understood to be dynamic, fluxing, with emergent, combinatorial and tectonic behaviour – it became instrumental as critique of the continuous first digital turn, through what has now emerged as the discrete. This has demonstrated that innovation occurred not in the translation between the two-dimensional drawing and building, but in the processes of feedback between tools, methods and a more critical meta-framework of digital thinking.

Notes

- 1 Sanford Kwinter, 'Landscapes of Change: Boccioni's "Stati d'animo" as a General Theory of Models', *Assemblage* 19 (1992): 52.
- 2 Mario Carpo, ed., *The Digital Turn in Architecture: 1992–2012* (London: John Wiley, 2012).
- 3 Carpo, *Digital Turn*.
- 4 Mario Carpo, *The Second Digital Turn: Design Beyond Intelligence* (Cambridge, MA: MIT Press, 2017), 7.
- 5 Gilles Retsin, ed., *Discrete: Reappraising the Digital in Architecture* (Chichester, UK: John Wiley, 2019).

- 6 Robin Evans, 'Translations from Drawing to Building', in Robin Evans, *Translations from Drawing to Building and Other Essays* (London: Architectural Association Publications, 1997), 153–93.
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